

A Network Gain Theorem for Interconnected Systems

Damián Marelli^{*,**} Minyue Fu^{***}

^{*} School of Automation, Guangdong University of Technology, Guangzhou 510006, China. Email: damianenviena@gmail.com

^{**} French-Argentinean International Center for Information and Systems Sciences, National Scientific and Technical Research Council, Rosario 2000, Argentina.

^{***} School of System Design and Intelligent Manufacturing, Southern University of Science and Technology, Shenzhen, China. Email: fumy@sustech.edu.cn

Abstract: This paper provides general network gain conditions for the input-to-state (ISS) stability of an interconnected nonlinear system. The result generalizes the well-known small-gain theorem for a single-loop system to networked systems under very general assumptions on individual sub-systems. The work also generalizes the previously known matrix small-gain theorem and cyclic small-gain theorem for networked systems.

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1. INTRODUCTION

The celebrated small-gain theorem is a pivotal tool in stability analysis and control design for single-loop nonlinear systems (Khalil, 2002). The theorem provides conditions on the gain bounds of the two sub-systems to guarantee stability when arranged in a feedback connection. The first version was introduced by G. Zames for linear gain bounds (Zames, 1966; Desoer and Vidyasagar, 2009), and extended to nonlinear bounds in (Hill, 1991; Mareels, 1992). These works were derived using the bounded-input, bounded-output (BIBO) stability concept. The more general input-to-state stability (ISS) concept, introduced for general dynamical systems by E. Sontag in (Sontag, 1989), was used in (Jiang et al., 1994, 1996), resulting in the so-called ISS small-gain theorem. Many other versions of the small-gain theorem are available in the literature.

A more general setup consists of a network of an arbitrary number of interconnected sub-systems, as depicted in Fig. 1. A survey of different versions of the ISS small-gain theorem for this setup is available in (Jiang and Liu, 2018). These methods can be classified according to the condition used to characterize the stability of each sub-system. This leads to two approaches. The first one is known as the *trajectory-based* condition, and consists in characterizing the stability of each sub-system using the ISS criterion (Liu et al., 2018). The second approach, known as the *Lyapunov-based* condition, characterizing the stability of each sub-system using ISS-Lyapunov functions (Liu et al., 2011, 2012). Classification can also be done according to the condition used to guarantee stability of the whole system. These conditions are stated in terms of the so-called *local gain functions*, which determines a gain associated to the link between every pair of intercon-

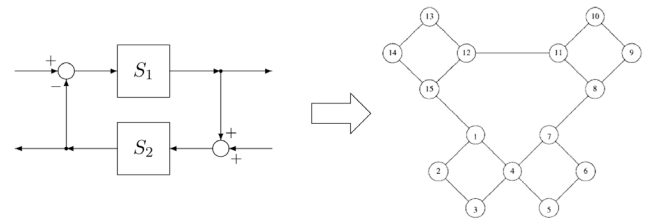


Fig. 1. Generalization of the small-gain theorem to networked systems

ected sub-systems. In the first group, often referred to as the *matrix small-gain theorem*, stability is guaranteed in terms of a property of a matrix of (possibly nonlinear) local gain functions (Dashkovskiy et al., 2007, 2010). The second one, called the *cyclic small-gain theorem*, asserts stability in terms of the composed gain associated to each simple loop formed by interconnected sub-systems (Liu et al., 2018, 2011, 2012).

In this paper, we report a new result that generalizes the small-gain theorem to a networked system. We follow the line of Lyapunov-based conditions, and derive a new result which is more general than both the matrix small-gain theorem and the cyclic small-gain theorem. Our result assesses the input-to-state (ISS) stability for a networked system in terms of a quantity which we call the *network gain*. We thus call our result the *network gain theorem*. The result holds under very general assumptions on the individual sub-systems, and only depends on the local gain functions of the sub-systems.

The rest of the paper is organized as follows. In Section 2 we formulate the problem. We derive our network gain theorem in Section 3. In Section 4 we compare the network

gain theorem with the matrix and cyclic small-gain theorems. In Section 5 we give some brief concluding remarks.

2. PROBLEM FORMULATION

In this section we describe the research problem addressed in this work. In Section 2.1 we introduce some required definitions and notation, and in Section 2.2 we formulate the research problem.

2.1 Definitions and notation

We use $\mathbb{N}$, $\mathbb{R}$, $\mathbb{R}_{\geq 0}$ and $\bar{\mathbb{R}}_{\geq 0}$ to denote the sets of natural, real, non-negative real and non-negative extended real numbers, respectively. For $I \in \mathbb{N}$ we use $\llbracket I \rrbracket = \{1, \dots, I\}$. For an I -dimensional vector x , both x_i and $[x]_i$ denote its i -th entry as well as $x_{\mathcal{I}}$ and $[x]_{\mathcal{I}}$ denote the vector formed by stacking the entries in the index set $\mathcal{I} \subseteq \llbracket I \rrbracket$. The 2-norm of a vector $x \in \mathbb{R}^I$ is denoted by $\|x\|$ and for a function $u : \mathbb{R} \rightarrow \mathbb{R}^I$, we define

$$\|u\|_{\infty} = \operatorname{ess\,sup}_{t \in \mathbb{R}_{\geq 0}} \|u(t)\|.$$

The distance $\delta(x, \mathcal{S})$ between a vector $x \in \mathbb{R}^I$ and a set $\mathcal{S} \subseteq \mathbb{R}^I$ is defined by $\delta(x, \mathcal{S}) = \inf_{y \in \mathcal{S}} \|x - y\|$. For a set $\mathcal{S}$, $\mathcal{S}^c$ is its (absolute) complement, $\mathcal{S} \setminus \mathcal{T}$ its relative complement with respect to set $\mathcal{T}$, $|\mathcal{S}|$ its number of elements and $2^{\mathcal{S}}$ its power set (i.e., the set of all subsets of $\mathcal{S}$). We use $\emptyset$ to denote the empty set, $\bar{\mathcal{B}}(a, r) = \{x : \|x - a\| \leq r\}$ to denote the closed ball of radius r centered in a , and $\bar{\mathcal{B}}(r) = \bar{\mathcal{B}}(0, r)$.

For a set $\mathcal{S} \subseteq \bar{\mathbb{R}}_{\geq 0}^I$, its supremum $\sup(\mathcal{S}) \in \bar{\mathbb{R}}_{\geq 0}^I$ and infimum $\inf(\mathcal{S}) \in \bar{\mathbb{R}}_{\geq 0}^I$ are defined by

$$\begin{aligned} [\sup(\mathcal{S})]_i &= \sup \{v_i : v \in \mathcal{S}\}, \\ [\inf(\mathcal{S})]_i &= \inf \{v_i : v \in \mathcal{S}\}. \end{aligned}$$

We use $\leq$ to denote the non-strict partial order on $\bar{\mathbb{R}}_{\geq 0}^N$ defined by $x \leq y$ if $x_n \leq y_n$, for all $n \in \llbracket N \rrbracket$, and $<$ to denote the strict partial order corresponding to $\leq$, i.e., $x < y$ if $x \leq y$ and $x \neq y$. We also use $x \prec y$ to denote $x_n < y_n$ for all $n \in \llbracket N \rrbracket$.

The classic definitions of positive definiteness as well as classes $\mathcal{K}$, $\mathcal{K}_{\infty}$ and $\mathcal{KL}$, for scalar valued functions (Khalil, 2002), naturally extend to vector valued functions using the partial order. We formally state this below.

Definition 1. We say that:

- A function $f : \bar{\mathbb{R}}^N \rightarrow \bar{\mathbb{R}}^M$ is positive semi-definite if: 1) $f(0) = 0$, and 2) $f(x) \geq 0$ for all $x \neq 0$.
- A function $f : \bar{\mathbb{R}}^N \rightarrow \bar{\mathbb{R}}^M$ is positive definite if 1) $f(0) = 0$, and 2) $f(x) > 0$ for all $x \neq 0$.
- A function $\alpha : \bar{\mathbb{R}}_{\geq 0}^N \rightarrow \bar{\mathbb{R}}_{\geq 0}^M$ belongs to class $\mathcal{K}(N, M)$ if 1) α is continuous, 2) α is strictly increasing (i.e., $x > y \Rightarrow \alpha(x) > \alpha(y)$), and 3) $\alpha(0) = 0$.
- A function α belongs to class $\mathcal{K}_{\infty}(N, M)$ if 1) $\alpha \in \mathcal{K}(N, M)$, and 2) $\lim_{\|x\| \rightarrow \infty} \|\alpha(x)\| = \infty$.
- A function $\beta : \bar{\mathbb{R}}_{\geq 0}^N \times \bar{\mathbb{R}}_{\geq 0} \rightarrow \bar{\mathbb{R}}_{\geq 0}^M$ belongs to class $\mathcal{KL}(N, M)$ if 1) β is continuous, 2) for each t , the map $s \mapsto \beta(s, t) \in \mathcal{K}(N, M)$, 3) for each s , the map $t \mapsto \beta(s, t)$ is decreasing, and 4) $\lim_{t \rightarrow \infty} \beta(s, t) = 0$.

For simplicity, we denote $\mathcal{K} = \mathcal{K}(1, 1)$, $\mathcal{K}_{\infty} = \mathcal{K}_{\infty}(1, 1)$ and $\mathcal{KL} = \mathcal{KL}(1, 1)$.

Definition 2. Let $f : \mathbb{R}^N \times \mathbb{R}^M \rightarrow \mathbb{R}^N$ be locally Lipschitz, and $u : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^M$ be measurable and locally essentially bounded. Let also $x_0 \in \mathbb{R}^N$ and $x : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^N$ be the unique solution of the following ordinary differential equation:

$$\dot{x}(t) = f(x(t), u(t)), \quad (1)$$

$$x(0) = x_0. \quad (2)$$

We say that the system (1)-(2) is input-to-state stable (ISS) if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that, for every x_0 and u ,

$$\|x(t)\| \leq \beta(\|x_0\|, t) + \gamma(\|u\|_{\infty}).$$

2.2 Research problem

We start by introducing networked control systems. Consider a graph $\mathcal{G}(\llbracket I \rrbracket, \mathcal{E})$ formed by a set $\llbracket I \rrbracket$ of nodes and a set of edges $\mathcal{E} \subset \llbracket I \rrbracket \times \llbracket I \rrbracket$. The open and closed neighborhood sets of Node i are defined by $\mathcal{N}_i = \{j \in \llbracket I \rrbracket : e_{i,j} \in \mathcal{E}\}$ and $\bar{\mathcal{N}}_i = \mathcal{N}_i \cup \{i\}$, respectively. Each node $i \in \llbracket I \rrbracket$ is associated with an ordinary differential equation

$$\dot{x}_i(t) = f_i(x_j(t) : j \in \bar{\mathcal{N}}_i, u_i(t)), \quad (3)$$

$$x_i(0) = x_{i,0}, \quad (4)$$

where $x_{i,0} \in \mathbb{R}^{N_i}$ is the initial condition, $f_i : \mathbb{R}^{\sum_{j \in \mathcal{N}_i} N_j} \times \mathbb{R}^{M_i} \rightarrow \mathbb{R}^{N_i}$ are locally Lipschitz with $f_i(0, 0) = 0$, and $u_i : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{M_i}$ are measurable and locally essentially bounded.

Let $N = \sum_{i=1}^I N_i$ and $M = \sum_{i=1}^I M_i$. Let also $x_0^{\top} = [x_{1,0}^{\top}, \dots, x_{I,0}^{\top}]$, $u^{\top} = [u_1^{\top}, \dots, u_I^{\top}]$ and $f : \mathbb{R}^N \times \mathbb{R}^M \rightarrow \mathbb{R}^N$ be defined by

$$[f(x, u)]_i = f_i(x_{\bar{\mathcal{N}}_i}, u_i), \quad (5)$$

where, with some abuse of notation, we use $x_{\bar{\mathcal{N}}_i}$ to denote the vector obtained by stacking the sub-vectors of x corresponding to the nodes with indices in $\bar{\mathcal{N}}_i$. Then, (3)-(4) can be compactly written as

$$\dot{x}(t) = f(x(t), u(t)), \quad (6)$$

$$x(0) = x_0. \quad (7)$$

In view of the above, we introduce the following definition of a networked system.

Definition 3. A networked system is a pair $(\mathcal{G}, f)$ where $\mathcal{G}$ is a graph and $f : \mathbb{R}^N \times \mathbb{R}^M \rightarrow \mathbb{R}^N$ is locally Lipschitz and satisfies (5).

The research problem consists in giving conditions on $\mathcal{G}$ and f to guarantee that the networked system $(\mathcal{G}, f)$ is ISS.

3. THE NETWORK GAIN THEOREM

In this section we introduce our network gain theorem. We first introduce the concept of *network Lyapunov triplet*.

Definition 4. Let $(\mathcal{G}, f)$ be a networked system. A networked Lyapunov triplet (NLT) (V, ϕ, δ) for $(\mathcal{G}, f)$ is formed by a vector of local pseudo-energy functions $V : \mathbb{R}^N \rightarrow \bar{\mathbb{R}}_{\geq 0}^I$, a vector of local gain functions $\phi : \bar{\mathbb{R}}_{\geq 0}^I \times \bar{\mathbb{R}}_{\geq 0}^I \rightarrow \bar{\mathbb{R}}_{\geq 0}^I$, and a function $\delta : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$, satisfying:

- (1) For each $i \in \llbracket I \rrbracket$, $[V(x)]_i = V_i(x_i)$, with $V_i : \mathbb{R}^{N_i} \rightarrow \mathbb{R}_{\geq 0}$ being continuously differentiable. Also, there exist $\underline{\alpha}, \bar{\alpha} \in \mathcal{K}_{\infty}$ such that, for all $x_i \in \mathbb{R}^{N_i}$

$$\underline{\alpha}(\|x_i\|) \leq V_i(x_i) \leq \bar{\alpha}(\|x_i\|). \quad (8)$$
- (2) For each $i \in \llbracket I \rrbracket$, $[\phi(v, \mu)]_i = \phi_i(v_{N_i}, \mu_i)$, with $\phi_i : \mathbb{R}_{\geq 0}^{N_i} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ being such that $v_{N_i} \mapsto \phi_i(v_{N_i}, \mu_i) \in \mathcal{K}(|N_i|, 1)$ for all $\mu_i \in \mathbb{R}_{\geq 0}$, and $\mu_i \mapsto \phi_i(v_{N_i}, \mu_i)$ is increasing for all v_{N_i} .
- (3) The map $\delta : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is continuous and positive definite. Also, for all $x_i \in \mathbb{R}^{N_i}$ and $u_i \in \mathbb{R}^{M_i}$,

$$V_i(x_i) \geq \phi_i(V_{N_i}(x), \|u_i\|) \Rightarrow \nabla^\top V_i(x_i) f_i(x_{N_i}, u_i) \leq -\delta(V_i(x_i)). \quad (9)$$

Definition 5. For an NLT (V, ϕ, δ) we define its compatible family $\mathcal{F}(V, \phi, \delta)$ as the set of all networked systems $(\mathcal{G}, f)$ for which (V, ϕ, δ) is an NLT. We say that $\mathcal{F}(V, \phi, \delta)$ is ISS if all its members are ISS, and that it is not ISS if at least one of its members is not ISS.

The vector function $V : \mathbb{R}^N \rightarrow \mathbb{R}_{\geq 0}^I$ assigns to each state x , the vector $V(x)$, which has one component per node. We call $V(x)$ the pseudo-energy vector associated with x . Applying condition (9), it follows that $V(x)$ is attracted by the set of pseudo-energy vectors $v \in \mathbb{R}_{\geq 0}^I$ satisfying the following condition:

$$v_i < \phi_i(v_{N_i}, \|u_i\|_{\infty}), \text{ for all } i \in \llbracket I \rrbracket. \quad (10)$$

This leads to the notion of *network attraction set* and *network gain* below.

Definition 6. Let (V, ϕ, δ) be a networked Lyapunov triplet. For each $\mu \in \mathbb{R}_{\geq 0}^I$, the network attraction set $\mathcal{Z}(\mu)$ is defined by

$$\mathcal{Z}(\mu) = \{v \in \mathbb{R}_{\geq 0}^I : v < \phi(v, \mu)\}. \quad (11)$$

Also, the network bound $\bar{z}(\mu) \in \mathbb{R}_{\geq 0}^I$ is defined by

$$\bar{z}(\mu) = \sup(\mathcal{Z}(\mu)), \quad (12)$$

and the *network gain* $\rho(\mu) \in \mathbb{R}_{\geq 0}$ by

$$\rho(\mu) = \|\bar{z}(\mu)\|. \quad (13)$$

We see from the above that the network gain only depends on the local gain functions of the sub-systems. The following example illustrates the above definitions using a simple two-node networked system.

Example 7. Consider the local gain functions shown in Figure 2. We see that the network attraction set is formed by two disconnected pieces. Because of the strict inequality in the definition of $\mathcal{Z}(\mu)$, this set does not include the segment departing from the second piece, on which $v = \phi(v)$. Hence, the network bound $\bar{z}(\mu)$ is at the beginning of that segment. The network gain $\rho(\mu)$ is also shown in the figure.

We can now state our main network stability result.

Theorem 8. [Network gain theorem] Let (V, ϕ, δ) be an NLT. Then, $\mathcal{F}(V, \phi, \delta)$ is ISS if $\rho : \mathbb{R}_{\geq 0}^I \rightarrow \mathbb{R}_{\geq 0}$ is positive semi-definite.

The rest of this section is devoted to show Theorem 8.

In view of (8), to study the stability of a networked system we could in principle study that of the pseudo-energy vector $V(x(t))$. There is however a difficulty in

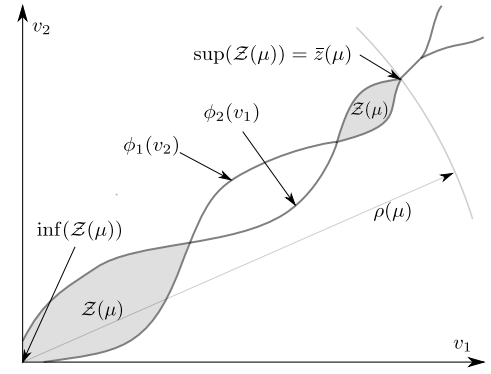


Fig. 2. Network attraction set, bound and gain of a two-node networked system example.

doing so. Suppose that the values of all $x_j(t)$, for $j \neq i$, are fixed. Then, (9) asserts that $V_i(x_i(t))$ decreases when $V(x(t))$ is outside the network attraction set $\mathcal{Z}(\mu)$; and therefore, $V(x(t))$ converges to that set. However, the same conclusion cannot be drawn if all $x_j(t)$, $j \neq i$, are let to evolve with time according to (3). This means that we cannot directly use $V(x(t))$ to show the stability of the networked system (3)-(4). To tackle this, we use V to define a generalized Lyapunov function (or vector Lyapunov function) $\mathcal{V}$, i.e., a vector-valued function which is strictly decreasing whenever it is non-zero. More precisely:

Definition 9. A map $\mathcal{V} : \mathbb{R}^N \rightarrow \mathbb{R}_{\geq 0}^I$ is called a generalized Lyapunov function for a networked system $(\mathcal{G}, f)$ if it is positive semi-definite, continuous and strictly decreasing (in the partial order defined on $\mathbb{R}_{\geq 0}^I$), i.e.,

$$\mathcal{V}(x(t)) > 0 \Rightarrow \mathcal{V}(x(s)) < \mathcal{V}(x(t)), \quad \forall s > t.$$

We now start the process of building a generalized Lyapunov function. Let $\mathcal{I} \subseteq \llbracket I \rrbracket$ denote a set of indices and $\text{index}_{\mathcal{I}}(i)$ denote the index of the element of $\mathcal{I}$ equal to $i \in \mathcal{I}$. The inclusion map $\iota_{\mathcal{I}} : \mathbb{R}_{\geq 0}^{|\mathcal{I}|} \rightarrow \mathbb{R}_{\geq 0}^I$ is defined by $[\iota_{\mathcal{I}}(w)]_i = w_{\text{index}_{\mathcal{I}}(i)}$, if $i \in \mathcal{I}$, and 0 otherwise. The projection map $\pi_{\mathcal{I}} : \mathbb{R}_{\geq 0}^I \rightarrow \mathbb{R}_{\geq 0}^{|\mathcal{I}|}$ by $[\pi_{\mathcal{I}}(v)]_i = v_i$, if $i \in \mathcal{I}$ and 0 otherwise. In words, the inclusion map $\iota_{\mathcal{I}}(w)$ yields a vector of I components, with the entries of vector w assigned to the indices in $\mathcal{I}$ and zero otherwise. Also, the projection map $\pi_{\mathcal{I}}(v)$ yields a vector having the components of v at the indices in $\mathcal{I}$ and zero otherwise.

For every $v \in \mathbb{R}_{\geq 0}^I$, $\mathcal{I} \subseteq \llbracket I \rrbracket$ and $w \in \mathbb{R}_{\geq 0}^{|\mathcal{I}|}$, we define the mixed vector $\mathbf{m}(v, \mathcal{I}, w) \in \mathbb{R}_{\geq 0}^I$ by

$$\mathbf{m}(v, \mathcal{I}, w) = \pi_{\mathcal{I}^c}(v) + \iota_{\mathcal{I}}(w).$$

i.e., $\mathbf{m}(v, \mathcal{I}, w) \in \mathbb{R}_{\geq 0}^I$ takes the vector v and replaces the components in the index set $\mathcal{I}$ by those from the vector w . Now, for each $\mu \in \mathbb{R}_{\geq 0}^I$, let $\mathbf{w}(v, \mathcal{I}, \mu) \in \mathbb{R}_{\geq 0}^{|\mathcal{I}|}$ be defined by

$$\mathbf{w}(v, \mathcal{I}, \mu) = \sup \left\{ w \in \mathbb{R}_{\geq 0}^{|\mathcal{I}|} : w \prec [\phi(\mathbf{m}(v, \mathcal{I}, w), \mu)]_{\mathcal{I}} \right\}.$$

In words, $\mathbf{w}(v, \mathcal{I}, \mu)$ is supremum of the set of vectors $w \in \mathbb{R}_{\geq 0}^{|\mathcal{I}|}$ satisfying the network attraction condition $w_{\text{index}_{\mathcal{I}}(i)} \prec \phi_i(\mathbf{m}(v, \mathcal{I}, w), \mu)$, for all $i \in \mathcal{I}$, while keeping constant the value of those entries of v corresponding to indices outside $\mathcal{I}$. We can then form a mixed vector $\mathbf{u}(v, \mathcal{I}, \mu) \in \mathbb{R}_{\geq 0}^I$ by replacing the components of v , with indices in $\mathcal{I}$, by those from $\mathbf{w}(v, \mathcal{I}, \mu)$, i.e.,

$$\mathbf{u}(v, \mathcal{I}, \mu) = \mathbf{m}(v, \mathcal{I}, \mathbf{w}(v, \mathcal{I}, \mu)).$$

In words, $\mathbf{u}(v, \mathcal{I}, \mu)$ is the vector obtained by keeping all entries of v corresponding to indices in $\mathcal{I}$ and maximizing all other components so as to arrive to the boundary of the network attraction condition. We therefore refer to $\mathbf{u}(v, \mathcal{I}, \mu)$ as a maximally mixed vector.

Now, for each $v, \mu \in \bar{\mathbb{R}}_{\geq 0}^I$, we define the vector $\mathbf{v}(v, \mu) \in \bar{\mathbb{R}}_{\geq 0}^I$ by

$$\mathbf{v}_i(v, \mu) = \max_{\mathcal{I} \in 2^{[I]}} \mathbf{u}_i(v, \mathcal{I}, \mu), \text{ for all } i \in [I],$$

and the set of indices $\mathcal{J}(v, \mu) \subset [I]$ by

$$\mathcal{J}(v, \mu) = \{i \in [I] : v_i < \mathbf{v}_i(v, \mu)\}.$$

In words, $\mathbf{v}(v, \mu)$ is defined such that each component $\mathbf{v}_i(v, \mu)$ equals the maximum among all possible maximally mixed vectors. Also, $\mathcal{J}(v, \mu)$ contains the indices i for which the maximized entry $\mathbf{v}_i(v, \mu)$ is greater than the corresponding non-maximized entry v_i of v .

The following lemma gives some instrumental properties of $\mathbf{v}$ and $\mathcal{J}$.

Lemma 10. For all $v, \mu \in \bar{\mathbb{R}}_{\geq 0}^I$:

- (1) $\mathbf{v}(v, \mu) \geq v$;
- (2) $\mathbf{v}(0, \mu) = \bar{z}(\mu)$;
- (3) $\mathbf{v}(v, \mu) \geq \bar{z}(\mu)$;
- (4) $\mathbf{v}(v, \mu) = \mathbf{u}(v, \mathcal{J}(v, \mu), \mu)$;
- (5) if $\mathbf{v}(v, \mu) > \bar{z}(\mu)$, then $[I] \setminus \mathcal{J}(v, \mu) \neq \emptyset$;
- (6) if $i \in [I] \setminus \mathcal{J}(v, \mu)$, then $v_i \geq \phi_i(v_{\mathcal{N}_i}, \mu)$.

Proof. We show each step separately.

1) It follows immediately since $\mathbf{v}(v, \mu) \geq \mathbf{u}(v, \emptyset, \mu) = v$.

2) For any $i \in [I]$,

$$\mathbf{v}_i(0, \mu) = \max_{\mathcal{I} \in 2^{[I]}} \mathbf{u}_i(0, \mathcal{I}, \mu) \geq \mathbf{u}_i(0, [I], \mu) = \bar{z}_i(\mu), \quad (14)$$

and

$$\begin{aligned} \mathbf{v}_i(0, \mu) &= \max_{\mathcal{I} \in 2^{[I]}} \mathbf{m}_i(0, \mathcal{I}, \mathbf{w}(0, \mathcal{I}, \mu)) \\ &\leq \max_{\mathcal{I} \in 2^{[I]}} \mathbf{m}_i(\bar{z}(\mu), \mathcal{I}, \mathbf{w}(\bar{z}(\mu), \mathcal{I}, \mu)) = \bar{z}_i(\mu). \end{aligned} \quad (15)$$

The claim then follows from (14) and (15).

3) Using a similar argument we obtain

$$\begin{aligned} \mathbf{v}(v, \mu) &\geq \mathbf{u}(v, [I], \mu) \\ &= \sup \{w \in \bar{\mathbb{R}}_{\geq 0}^I : w \prec \phi(w, \mu)\} = \bar{z}(\mu). \end{aligned}$$

4) Let $\mathcal{C} \subset [I]$. We have $\mathcal{C} = \mathcal{A} \cup \mathcal{B}$, $\mathcal{A} = \mathcal{J}(v, \mu) \cap \mathcal{C}$ and $\mathcal{B} = \mathcal{J}^c(v, \mu) \cap \mathcal{C}$. Now, for every $i \in \mathcal{J}(v, \mu)$, $v_i < \mathbf{v}_i(v, \mu)$. Then, since $\mathcal{A} \subseteq \mathcal{J}(v, \mu)$,

$$\begin{aligned} \mathbf{u}(v, \mathcal{J}(v, \mu), \mu) &= \pi_{\mathcal{J}(v, \mu)} \mathbf{c}(v) + \iota_{\mathcal{J}(v, \mu)}(\mathbf{w}(v, \mathcal{J}(v, \mu), \mu)) \\ &> \pi_{\mathcal{A}} \mathbf{c}(v) + \iota_{\mathcal{A}}(\mathbf{w}(v, \mathcal{A}, \mu)) = \mathbf{u}(v, \mathcal{A}, \mu). \end{aligned} \quad (16)$$

Also, for every $i \notin \mathcal{J}(v, \mu)$,

$$v_i = \mathbf{v}_i(v, \mu) > \max_{i \in \mathcal{I} \in 2^{[I]}} \mathbf{u}_i(v, \mathcal{I}, \mu).$$

Then, since $\mathcal{B} \subseteq \mathcal{J}^c(v, \mu)$,

$$\begin{aligned} \mathbf{u}(v, \mathcal{A}, \mu) &= \pi_{\mathcal{A}} \mathbf{c}(v) + \iota_{\mathcal{A}}(\mathbf{w}(v, \mathcal{A}, \mu)) \\ &> \pi_{\mathcal{C}} \mathbf{c}(v) + \iota_{\mathcal{C}}(\mathbf{w}(v, \mathcal{C}, \mu)) = \mathbf{u}(v, \mathcal{C}, \mu). \end{aligned} \quad (17)$$

From (16)-(17), for every $i \in [I]$, $\mathbf{u}_i(v, \mathcal{J}(v, \mu), \mu) \geq \mathbf{u}_i(v, \mathcal{C}, \mu)$, and the result follows since the choice $\mathcal{C}$ was arbitrary.

5) Suppose that $[I] \setminus \mathcal{J}(v, \mu) = \emptyset$, or equivalently, $\mathcal{J}(v, \mu) = [I]$. Then, from Property 4, $\mathbf{v}(v, \mu) = \mathbf{u}(v, [I], \mu) = \bar{z}(\mu)$, and the result follows.

6) If $i \in [I] \setminus \mathcal{J}(v, \mu)$, then

$$v_i = \mathbf{v}_i(v, \mu) \geq \mathbf{u}_i(v, \{i\}, \mu) = \phi_i(v_{\mathcal{N}_i}, \mu).$$

□

We can now use $\mathbf{v}$ to introduce a generalized Lyapunov function for the networked system $(\mathcal{G}, f)$.

Proposition 11. Let $(\mathcal{G}, f)$ be a networked system with NLT (V, ϕ, δ) and $\mu \in \bar{\mathbb{R}}_{\geq 0}^I$. If $\|u_i\|_{\infty} \leq \mu_i$, $i \in [I]$, then the map $\mathcal{V} : \mathbb{R}^N \rightarrow \bar{\mathbb{R}}_{\geq 0}^I$ defined by

$$\mathcal{V}(x) = \mathbf{v}(V(x), \mu) - \bar{z}(\mu),$$

is a generalized Lyapunov function for $(\mathcal{G}, f)$.

Proof. It readily follows from Lemma 10-2 and 3 that $\mathcal{V}$ is positive semi-definite and it is easy to see, from the continuity of ϕ , that it is continuous. Hence we need to show that

$$\mathcal{V}(x(t)) > 0 \Rightarrow \mathcal{V}(x(s)) < \mathcal{V}(x(t)), \quad \forall s > t.$$

If $\mathcal{V}(x(t)) > 0$, then $\mathbf{v}(V(x(t)), \mu) > \bar{z}(\mu)$. Then, from Lemma 10-5 and 6, for all $i \in [I] \setminus \mathcal{J}(V(x(t)), \mu) \neq \emptyset$,

$$V_i(x_i(t)) \geq \phi_i(V_{\mathcal{N}_i}(x(t)), \mu_i) \geq \phi_i(V_{\mathcal{N}_i}(x(t)), \|u_i(t)\|).$$

It then follows from (9) that

$$\begin{aligned} \dot{V}_i(x_i(t)) &= \nabla^\top V_i(x_i(t)) f_i(x_{\mathcal{N}_i}(t), u_i(t)) \\ &\leq -\delta(V_i(x_i(t))). \end{aligned} \quad (18)$$

Now, in view of Lemma 10-4, $\mathbf{v}_i(V(x(t)), \mu) = \mathbf{u}_i(V(x(t)), \mathcal{J}(V(x(t)), \mu), \mu) = V_i(x_i(t))$. Hence, all components $\mathbf{v}_i(V(x(t)), \mu)$, with $i \in [I] \setminus \mathcal{J}(V(x(t)), \mu)$, decrease. In view of the monotonicity of ϕ , all components $i \in \mathcal{J}(V(x(t)), \mu)$ will not increase, and the result follows. □

We can now show our main result.

Proof for Theorem 8 We split the proof into steps:

1) Let $\mu \in \bar{\mathbb{R}}_{\geq 0}^I$ be given by $\mu_i = \|u_i\|_{\infty}$, for all $i \in [I]$. Consider the rectangle set defined by

$$\mathcal{R}(z) = \{v \in \bar{\mathbb{R}}_{\geq 0}^I : v \leq z\},$$

and define $\mathcal{Q}(\mu) = V^{-1}(\mathcal{R}(\bar{z}(\mu)))$. From Lemma 10-1,

$$\begin{aligned} \mathcal{P}(\mu) &\triangleq \{x \in \mathbb{R}^N : \mathbf{v}(V(x), \mu) = \bar{z}(\mu)\} \\ &\subseteq \{x \in \mathbb{R}^N : V(x) \leq \bar{z}(\mu)\} = \mathcal{Q}(\mu). \end{aligned}$$

Also, from Lemma 10-4

$$\begin{aligned} \mathcal{Q}(\mu) &= \{x \in \mathbb{R}^N : V(x) \leq \bar{z}(\mu)\} \\ &\subseteq \{x \in \mathbb{R}^N : \mathcal{J}(V(x), \mu) = [I]\} \\ &= \{x \in \mathbb{R}^N : \mathbf{v}(V(x), \mu) = \bar{z}(\mu)\} = \mathcal{P}(\mu). \end{aligned}$$

Hence $\mathcal{P}(\mu) = \mathcal{Q}(\mu)$. Now, since $\mathcal{P}(\mu) = \{x \in \mathbb{R}^N : \mathcal{V}(x) = 0\}$, it follows from Proposition 11 that $\mathcal{P}(\mu)$ is an global ultimate bounding set for (V, ϕ) . We then have

$$\lim_{t \rightarrow \infty} d(x(t), \mathcal{Q}(\mu)) = 0. \quad (19)$$

2) Let

$$\bar{S}(t, \alpha) = \sup_{\|x_0\| = \alpha} d(x(t, x_0), \mathcal{Q}(\mu)). \quad (20)$$

where we used $x(t, x_0)$ to denote the explicit dependence of $x(t)$ on x_0 . We aim to show

$$\lim_{t \rightarrow \infty} \bar{S}(t, \alpha) = 0, \quad (21)$$

For each $\epsilon > 0$, let

$$\tau(x_0, \epsilon) = \sup \{t \in \mathbb{R}_{\geq 0} : d(x(t, x_0), \mathcal{Q}(\mu)) \geq \epsilon\}.$$

The meaning of $\tau(x_0, \epsilon)$ is that, if the networked system is started from x_0 , then $d(x(t, x_0), \mathcal{Q}(\mu)) \leq \epsilon$, for all $t \geq \tau(x_0, \epsilon)$. We then define the maximum of all these times $\tau(x_0, \epsilon)$ taken over all initial values x_0 having norm equal to α , i.e.,

$$\bar{\tau}(\epsilon) = \sup_{\|x_0\|=\alpha} \tau(x_0, \epsilon).$$

It then follows that (21) would hold if $\bar{\tau}(\epsilon) < \infty$ for all $\epsilon > 0$. To show this, suppose that $\bar{\tau}(\epsilon) = \infty$. We could then choose a sequence $x_{0,n}$, $n \in \mathbb{N}$, such that $\lim_{n \rightarrow \infty} \tau(x_{0,n}, \epsilon) = \infty$. Since the set $\{x_0 : \|x_0\| = \alpha\}$ is compact, there exists a sub-sequence x_{0,n_k} , $k \in \mathbb{N}$, which converges to a limit point $x_{0,*}$. But we must then have $\tau(x_{0,*}, \epsilon) = \infty$. This contradicts (19), hence $\bar{\tau}(\epsilon) < \infty$, and (21) follows. It is easy to see that (21) implies the existence of $\beta \in \mathcal{KL}$ such that $\bar{S}(t, \alpha) \leq \beta(\alpha, t)$. It then follows from (20) that

$$d(x(t), \mathcal{Q}(\mu)) \leq \beta(\|x_0\|, t).$$

3) If $\rho(\mu) > 0$, there exists $0 \neq v \in \mathcal{Z}(\mu)$. Since $\mathcal{Z}(\mu)$ is open, there exists $r > 0$ such that $\bar{\mathcal{B}}(v, r) \subset \mathcal{Z}(\mu)$. We then define

$$f_i(x_{\mathcal{N}_i}, u_i) = \tilde{f}_i(x_{\mathcal{N}_i}, u_i) \kappa_i(x_{\mathcal{N}_i}, v, r),$$

where $\kappa_i(\tilde{x}, v, r) = \frac{\|V_{\mathcal{N}_i}(\tilde{x}) - v_{\mathcal{N}_i}\|}{r}$ if $\|V_{\mathcal{N}_i}(\tilde{x}) - v_{\mathcal{N}_i}\| < r$ and 1 otherwise. Let $f : \mathbb{R}^N \times \mathbb{R}^M \rightarrow \mathbb{R}^N$ be defined by $[f(x, u)]_i = f_i(x_{\mathcal{N}_i}, u_i)$. It is easy to see that f is locally Lipschitz and satisfies (5). Hence, $(\mathcal{G}, f)$ is a networked system, and that (V, ϕ, δ) is an NLT for this system. Let $\xi = V^{-1}(v)$. Then, $f(\xi, u) = 0$, for all $u \in \mathbb{R}^M$. Since $v \neq 0$, then, from (8), $\xi \neq 0$, and therefore $(\mathcal{G}, f)$ is non-GAS.

4) If $\rho(\mu) = \infty$, choose $\delta > 0$. There exists $v_n \in \mathcal{Z}(\mu)$, $n \in \mathbb{N}$, such that $\lim_{n \rightarrow \infty} \|v_n\| = \infty$ and $\inf_{n \in \mathbb{N}} \|v_{n+1} - v_n\| > \delta$. We can also choose $r_n > 0$, $n \in \mathbb{N}$ such that $\bar{\mathcal{B}}(v_n, r_n) \subset \mathcal{Z}(\mu)$. We then define $f_i(x_{\mathcal{N}_i}, u_i) = \tilde{f}_i(x_{\mathcal{N}_i}, u_i) \sum_{n=1}^{\infty} \kappa_i(x_{\mathcal{N}_i}, v_n, r_n)$. Since $\delta > 0$, it is easy to see that f is locally Lipschitz. The rest of the argument follows the steps of the proof of part 1.

5) The direct part of the theorem is an immediate consequence of Step 2), by choosing $\mu_i = \|u_i\|_{\infty}$. For the converse part, since ρ is clearly increasing, if it is not positive semi-definite, then either $\rho(\mu) > 0$ for some $\mu \neq 0$ or $\rho(\mu)$ fails to be finite for some $\mu \in \mathbb{R}_{\geq 0}^I$. In the first case, Step 3) asserts the existence of a non-ISS networked system for which (V, ϕ, δ) is an NLT. In the second case, the same assertion follows from Step 4). $\square$

4. COMPARISONS

In this section, we compare Theorem 8 with the matrix small-gain theorem (Dashkovskiy et al., 2010) and the cyclic small-gain theorem (Liu et al., 2011). The main purpose is to prove that these results can be derived from Theorem 8.

We start with the matrix small-gain theorem.

Definition 12. Let $(\mathcal{G}, f)$ be a networked system with NLT (V, ϕ, δ) . We say that $\sigma \in \mathcal{K}_{\infty}^I$ is an Ω -path if, for each $i \in \llbracket I \rrbracket$, σ_i and σ_i^{-1} are locally Lipschitz and, for all $t \in \mathbb{R}_{\geq 0}$,

$$\phi(\sigma(t), 0) \prec \sigma(t).$$

Matrix Small-gain Theorem (Dashkovskiy et al., 2010, Theorem 5.3) Let $(\mathcal{G}, f)$ be a networked system with NLT (V, ϕ, δ) . Let ϕ be given by

$$\phi_i(v, \mu) = \lambda_i(\nu_{i,j}(v_j) : j \in \mathcal{N}_i, v_i(\|\mu\|)),$$

for all $i \in \llbracket I \rrbracket$, where $\lambda_i \in \mathcal{K}_{\infty}(|\mathcal{N}_i|, 1)$, $\nu_{i,j} \in \mathcal{K}_{\infty}$, $j \in \mathcal{N}_i$, and $v_i \in \mathcal{K}$. If matrix $A \in \mathbb{R}^{I \times I}$ defined by

$$[A]_{i,j} = \begin{cases} 1, & j \in \mathcal{N}_i, \\ 0, & j \notin \mathcal{N}_i. \end{cases}$$

is irreducible,

$$v \not\leq \phi(v, 0), \quad (22)$$

and there exists $\psi \in \mathcal{K}_{\infty}(1, I)$ such that, for all $t \in \mathbb{R}_{\geq 0}$,

$$\phi(\sigma(t), \psi(t)) \prec \sigma(t), \quad (23)$$

where σ is an Ω -path, then $(\mathcal{G}, f)$ is ISS.

Now we prove this result using Theorem 8.

Proof. We need to show that ρ is a positive definite function. It is easy to see that ρ is increasing. Also, condition (22) implies that $\mathcal{Z}(0) = \emptyset$ and therefore $\rho(0) = 0$. So we need to show that $\rho(\mu) \in \mathbb{R}$ for all $\mu \in \mathbb{R}_{\geq 0}^I$ (i.e., that $\rho(\mu)$ is finite for finite μ).

Let $\tilde{\phi}_i(v, r) = \lambda_i(\nu_{i,j}(v_j) : j \in \mathcal{N}_i, v_i(r))$, so that $\phi(v, \mu) = \tilde{\phi}(v, \|\mu\|)$. Let $s \in \mathbb{R}_{\geq 0}$ be such that $\|\psi(s)\| = \|\mu\|$. Then

$$\phi(\sigma(s), \psi(s)) = \tilde{\phi}(\sigma(s), \|\mu\|) = \phi(\sigma(s), \mu).$$

Suppose that there exists $\mu \in \mathbb{R}_{\geq 0}^I$ such that $\rho(\mu) = \infty$. Then, there exists $v(n) \in \mathcal{Z}(\mu)$ with $\lim_{n \rightarrow \infty} \|v(n)\| = \infty$. Let $\bar{v} = \min\{v(n) : v(n) \not\leq \sigma(s)\}$, and $\bar{t} = \inf\{t \in \mathbb{R}_{\geq 0} : \sigma(t) \geq \bar{v}\}$. In particular, there exists $j \in \llbracket I \rrbracket$ such that

$$\sigma_j(\bar{t}) = \bar{v}_j. \quad (24)$$

Clearly, $\bar{t} \geq s$. Hence,

$$\phi(\sigma(\bar{t}), \mu) = \phi(\sigma(\bar{t}), \psi(s)) \leq \phi(\sigma(\bar{t}), \psi(\bar{t})) \prec \sigma(\bar{t}). \quad (25)$$

But since $v_* \in \mathcal{Z}(\mu)$, then

$$\phi(v_*, \mu) > v_*. \quad (26)$$

In view of (24), equations (25) and (26) are incompatible. Hence $\rho(\mu)$ must be finite. $\square$

We next state the cyclic small-gain theorem.

Cyclic Small-gain Theorem (Liu et al., 2011, Theorem 1) Let $(\mathcal{G}, f)$ be a networked system with NLT (V, ϕ, δ) . Let ϕ be given by

$$\phi_i(v_{\mathcal{N}_i}, \mu) = \max\{\nu_{i,j}(v_j) : j \in \mathcal{N}_i, v_i(\mu_i)\}, \quad (27)$$

for $i \in \llbracket I \rrbracket$, where $v_i \in \mathcal{K}$ and $\nu_{i,j} \in \mathcal{K}$, $j \in \mathcal{N}_i$. If for every loop $(i_1, \dots, i_K, i_1)$, the map

$$\nu_{v_1, v_2} \circ \dots \circ \nu_{v_{K-1}, v_K} < \text{id}, \quad (28)$$

where $\text{id} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ denotes the identity map, i.e., $\text{id}(x) = x$, then $(\mathcal{G}, f)$ is ISS.

Again, we prove this result using Theorem 8.

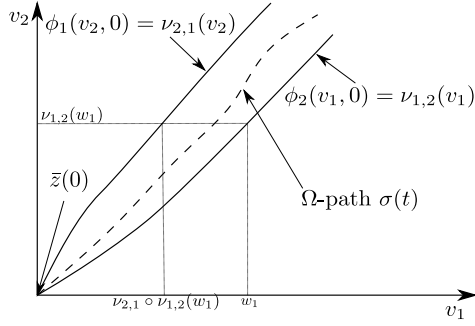


Fig. 3. Comparison of matrix and cyclic small-gain theorems with the proposed network gain theorem.

Proof. We need to show that ρ is bounded by a class $\mathcal{K}(I, 1)$ function. Clearly, ρ is increasing. So we need to show that $\rho(0) = 0$ and $\rho(\mu) \in \mathbb{R}$ for all $\mu \in \mathbb{R}_{\geq 0}^I$ (i.e., that $\rho(\mu)$ is finite for finite μ).

1) We show $\rho(0) = 0$. Let $v \in \mathcal{Z}(0)$. Choose $i_1 \in \llbracket I \rrbracket$ and generate a sequence of indices using $i_{k+1} = \arg \max_{j \in \mathcal{N}_i} \nu_{i_k, j}(v_j)$, until $i_{k+1} = i_1$. This produces a loop $(i_1, \dots, i_K, i_1)$. Since $v \in \mathcal{Z}(0)$, we must have $v_{i_k} < \nu_{i_k, i_{k+1}}(v_{i_{k+1}})$. We then have $v_{i_1} < \nu_{i_1, i_2} \circ \dots \circ \nu_{i_K, i_1}(v_{i_1})$. Hence, in view of (28), no $v_{i_1} \in \mathbb{R}_{\geq 0}$ can satisfy the above. We then obtain $\mathcal{Z}(0) = \emptyset$, thus $\rho(0) = 0$.

2) We show $\rho(\mu) \in \mathbb{R}$ for all $\mu \in \mathbb{R}_{\geq 0}^I$. Let $v = [v_1, \dots, v_I]^\top$. We have

$$\begin{aligned} \mathcal{Z}(\mu) &= \{v \in \mathbb{R}_{\geq 0}^I : v < \phi(v, \mu)\} \\ &= \mathcal{Z}(0) \cup \{v \in \mathbb{R}_{\geq 0}^I : v < v(\mu)\} = \{v \in \mathbb{R}_{\geq 0}^I : v < v(\mu)\}. \end{aligned}$$

It then follows that $\rho(\mu) = \|v(\mu)\|$ and $\rho(\mu) \in \mathbb{R}$ follows from v being class $\mathcal{K}(I, 1)$. $\square$

Finally, we use two examples to demonstrate the proposed network gain theorem.

Example 13. In this simple example we compare the proposed network gain theorem with the matrix small-gain theorem and the cyclic small-gain theorem. Consider a networked system with two nodes and local gain functions ϕ_1 and ϕ_2 , for the case $\mu = 0$, shown in Figure 3. The figure shows that $\bar{z}(0) = 0$, hence $\mathcal{Z}(0) = \emptyset$. The figure also shows that $\mathcal{Z}(0) = \emptyset$ implies (22) and the existence of an Ω -path, as required by the matrix small-gain theorem. We also see that $\mathcal{Z}(0) = \emptyset$ implies that the composition $\nu_{1,2} \circ \nu_{2,1} < \text{id}$, as well as $\nu_{2,1} \circ \nu_{1,2} < \text{id}$, although the latter is not explicitly shown in the figure, as required by the cyclic small-gain theorem. All these properties hold in the case $\mu = 0$, i.e., when the input is zero. The essential difference between these theorems is in how the case $\mu \neq 0$ is dealt with. The matrix and cyclic small-gain theorems do so by imposing structural assumptions on ϕ_1 and ϕ_2 , and the former further imposes condition (23). In contrast, the proposed network gain theorem directly states the condition on the network gain $\rho(\mu)$.

Example 14. Finally, we consider a networked system with $I = 10$ nodes, connected such that its connection graph has a ring topology, i.e., the state of node i acts as input for nodes $i - 1 \bmod I$ and $i + 1 \bmod I$, where $\bmod$ denotes the modulo operation. For each $i \in \llbracket I \rrbracket$, the dynamics of node i is given by

$$\dot{x}_i(t) = \text{sgn}(x_i(t)) \phi_i(x_{i-1 \bmod I}^2(t), x_{i+1 \bmod I}^2(t)) - [|x_i(t)| + 1] x_i(t)$$

$$\phi_i(v_{i-1}, v_{i+1}) = \sqrt{\alpha_{i,i-1 \bmod I} v_{i-1 \bmod I} + \alpha_{i,i+1 \bmod I} v_{i+1 \bmod I}},$$

and $\alpha_{i,i-1 \bmod I} = \alpha_{i,i+1 \bmod I} = 0.5$. We choose $V_i(x_i) = x_i^2$ for all $i \in \llbracket I \rrbracket$. Using the fact that nodes are homogeneous, we can show that the network gain is $\rho(\mu) = (1 + \sqrt{1 + 4\mu})/2$, hence, the networked system is ISS.

5. CONCLUSIONS

We have proposed a new result for assessing stability in a network of interconnected sub-systems. Our result, called the network gain theorem, generalizes the previously known matrix small-gain theorem and cyclic small-gain theorem. The network gain only depends on the local gain functions of the sub-systems. Examples have been provided to illustrate the network gain theorem. A related but important problem is how to test the network gain conditions. We will report the work on this problem separately.

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